MILITARY SPECIFICATION

COOLER, CRYOGENIC, MECHANICAL HD-1033B/UA and HD-1033C/UA

This specification is approved for use by USACECOM, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module with 2 interchangeable models; Cooler, Cryogenic, Mechanical HD-1033B/UA and HD-1033C/UA.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. In the event of a conflict between the document referenced herein and the contents of this specification, the contents of this specification shall be considered a superceding requirement.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-704	- Electrical Power, Aircraft Characteristics and Utilization of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications - Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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MIL-STD-726 - Packaging Requirement Codes
MIL-STD-810 - Environmental Test Methods

DRAWINGS

USA ELECTRONICS R&D COMMAND

SM-D-969477 - Cooler, Cryogenic, Mechanical HD-1033B/UA
SM-D-971500 - Cooler, Cryogenic, Mechanical HD-1033C/UA
54490-5004372 - Cryogenic Heat Load Assembly (1 Watt Integral)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

ANSI S1.11-1971 - Specification for Octave, Half-Octave, and Third-Octave Band Filter Sets

(Application for copies of ANSI standards should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Description. The Cooler, Cryogenic, Mechanical HD-1033B/UA and HD-1033C/UA, referred to herein as the cooler, is a closed-cycle refrigerator for cooling the infrared detector and interfacing with the Dewar of the infrared detector-Dewar package in an infrared system. The two models differ in their internal configuration but are identical in performance and external configuration. In system usage they are completely interchangeable.

3.2 Construction. The cooler shall be constructed in accordance with SM-D-969477 or SM-D-971500 and as specified herein.

3.2.1 Weight. The weight of the cooler shall be 3.7, (+0.1, -0.2) pounds.

3.3 First article. The contractor shall furnish first article assemblies in accordance with 4.3.

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings. Materials, parts, and processes not specified shall be selected by the contractor in accordance with MIL-P-11268.

3.5 Components. The cooler shall consist of an ac motor-driven, helium-filled compressor with a coldfinger.

3.6 Performance characteristics. The cooler shall meet the performance characteristics specified herein when adequate heat sinking or convective cooling is

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provided to ensure that any point on the cooler cylinder head shall fall within 5°C to 15°C above ambient air temperature.

3.6.1 Cooling capacity. The cooler with thermal load (copper mass) shall provide the minimum refrigeration capacity at 80 K maximum as shown in figure 1, curve A.

3.6.2 Cooldown time. The cooldown time to reach a cold tip temperature of 80 K from ambient with a 1440, (+25, -0) joule copper thermal mass (IAW Drawing 54490-5004372) (from 300 K to 80 K) shall be in accordance with figure 2.

3.6.3 Input power. The total input power to the cooler shall not exceed that shown in figure 3 with 1440, (+25, -0) joule thermal mass after cold tip temperature has stabilized with the heat load shown in figure 1, curve A. The input voltage shall be 117, ±2 volts alternating current (VAC), 400, ±20 hertz (Hz) power source.

3.6.4 Acoustic noise. The cooler noise emission shall not be more than the values listed below, when measured at a distance of 3 feet.

Sound pressure values

Center frequency (Hz)	Octave band (Hz)	Maximum sound pressure level (dB), reference 0.0002 microbar
125	87-175	45
250	175-350	45
500	350-700	48
1000	700-1400	55
2000	1400-2800	60
4000	2800-5600	65
8000	5600-11200	65

3.6.5 Leak rate. The leak rate of the cooler shall not be greater than 2.5×10^{-6} cubic centimeters per second (cc/sec) helium equivalent at an ambient temperature of +23°C, ±5°C.

3.6.6 Vibration output. The maximum amplitude, at the ambient temperature of +23°C, ±5°C, of the vibrational force induced by the cooler at a frequency of 26, ±4 Hz or any of the next eight harmonics along the compressor piston and coldfinger axes shall not exceed 0.35 pound (semi-amplitude) and shall not exceed 0.5 pound (semi-amplitude) along the motor axis. The maximum amplitude of the torque induced by the cooler about the motor, piston and coldfinger axes at 26, ±4 Hz or any of the next eight harmonics shall not exceed:

a. 4.0 in-oz (semi-amplitude) compressor piston axis
b. 4.0 in-oz (semi-amplitude) coldfinger axis
c. 15.0 in-oz (semi-amplitude) motor axis.

3.6.7 Restart. The cooler shall restart after having cooled to the operating temperature of at least 80K at any ambient from - 54C to + 71C.

3.7 Environmental conditions.

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3.7.1 Temperature shock. The cooler shall not be damaged (see 6.3.1) by sudden changes in temperature between -62°C and +95°C.

3.7.2 High temperature. The cooler shall not be damaged by storage to +95°C or operation up to +71°C.

3.7.3 Low temperature. The cooler shall not be damaged by operation to -54°C or storage to -62°C.

3.7.4 Shock. The cooler shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The cooler shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 4.

3.8 Burn-in. Each cooler delivered shall have accumulated a minimum of 12 hours burn-in over the temperature profile in accordance with figure 5.

3.9 Reliability. The cooler shall have a lower mean-time-to failure (MTTF) of at least 1,000 hours.

3.10 Treatment. Unless otherwise specified, the cooler and its components and parts shall be cleaned and treated in accordance with the applicable drawing.

3.11 Nameplates and product marking. Unless otherwise specified, the cooler, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.10).

4.3 First article inspection. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

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4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and a 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Treatment not as specified	2.5 1/	3.10	
Marking not as specified	2.5	3.11	
Workmanship not as specified	2.5	3.12	MIL-STD-454
Dimensions not as specified		3.2	SM-D-969477 or SM-D-971500

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.

4.3.2 Test. Upon successful completion of the inspections specified in 4.3.1, four first article coolers shall be subjected to reliability testing and 2 first article coolers shall be subjected to all other tests listed in table II. Tests may be conducted in any order. Failure of any test shall constitute first article failure.

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.f).

TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Cooling capacity	3.6.1	4.6.1
Cooldown time	3.6.2	4.6.2
Input power	3.6.3	4.6.3
Acoustic noise	3.6.4	4.6.4
Leak rate	3.6.5	4.6.5
Vibration output	3.6.6	4.6.6
Restart	3.6.7	4.6.7
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3
Shock	3.7.4	4.7.4
Vibration	3.7.5	4.7.5
Reliability	3.9	4.9

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4.4 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the cooler shall be operated at an ambient temperature of +23°C, +5°C. Adequate heat sinking or convective cooling shall be provided to ensure that any point on the cooler cylinder head shall fall within 5°C to 15°C above ambient air temperature (see 3.6).

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1. Burn-in. Burn-in shall be in accordance with 4.8 and shall be performed upon each cooler that has passed the inspection of 4.5.1.

4.5.2.2 Group A inspection. Group A inspection shall be conducted on all coolers which have completed burn-in specified in 4.5.2.1. Group A tests listed in table III may be performed in any order unless otherwise specified. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Cooling capacity	3.6.1	4.6.1
Cooldown time	3.6.2	4.6.2
Input power	3.6.3	4.6.3
Leak rate	3.6.5	4.6.5
Restart	3.6.7	4.6.7

4.5.2.3 Group B inspection. Not required.

4.5.2.4 Group C inspection. Group C inspections shall be conducted on coolers selected from units which have passed the tests in 4.5.2.2 (see 6.2.h). The sample(s) shall be tested in accordance with the inspections listed in table IV. Samples shall be selected in accordance with 4.5.2.4.1. Group C tests listed in table IV may be performed in any order.

4.5.2.4.1 Sampling for group C inspection. Group C samples shall be selected as follows:

Monthly lot size	Number of sample units monthly
1-99	1
100-299	3
300-499	5
500-999	7
1000 or more	10

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Acoustic noise	3.6.4	4.6.4
Vibration output	3.6.6	4.6.6
Temperature shock	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.3	4.7.3

4.5.2.4.2 Group C failures. Actions required relative to group C failures shall be specified in the contract (see 6.2.c.(2)).

4.5.2.4.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to the tests of table III.

4.5.2.5 Group D inspection. Group D inspections shall be conducted on coolers selected from units which have passed the tests in 4.5.2.2. The samples shall be tested in accordance with 4.9.

4.5.2.5.1 Sampling for Group D inspection. The reliability acceptance test sample size shall be 4 units. The sample shall be randomly selected from units produced during the sampling period. The frequency of the test shall be as follows:

Monthly Lot Size	Frequency of Group D Inspection
1-9	None
10-49	Annual
50 or more	Semiannual

4.5.2.5.2 Group D Failures. Actions required relative to group D failures shall be specified in the contract (see 6.2.c.(3)).

4.5.2.5.3 Disposition of Group D Samples. Group D samples shall not be considered as part of the procurement quantities (see 6.2.f).

4.6 Test methods.

4.6.1 Cooling capacity. The cooling capacity shall be measured by attaching to the cooler a test Dewar and coldstation. The heat load shall be applied 30 minutes after achieving cooldown. Capacity shall be measured not less than 20 minutes later. The tests of paragraphs 4.6.1, 4.6.2 and 4.6.3 shall be performed using test equipment consisting of the following items to measure cooldown and refrigeration capacity:

1. Test dewar

2. Coldstation

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3. Appropriate power supplies and meters

4. High vacuum source

   a. The test dewar shall be made of either brass or aluminum with a highly polished surface finish of at least an 8 RMS. The test dewar in conjunction with the high vacuum source shall enclose the coldfinger and coldstation in a vacuum of $1x10^{-5}$torr or lower.

   b. The 1440 joule coldstation shall consist of the following elements as defined on drawing 54490 5004372: copper block, temperature sensor, resistor, thermally conductive epoxy and a clamp screw.

   c. The coldstation assembly will have the following characteristics:

1. The temperature sensor and resistor will be bonded to the copper block with the thermally conductive epoxy.

2. The thermal energy which must be removed when cooling the coldstation from 300 K to 80 K shall be 1440, (+25, -0) joules. The mass of the coldstation shall be adjusted until it meets this requirement.

3. The resistor in the coldstation shall have a voltage applied to obtain a 1000, (+20, -0) mw thermal load. The applied voltage shall be such that the product of the voltage and the current of the resistor shall equal 1000, (+20, -0) mw.

4. The resistor and temperature sensor shall be connected to a vacuum feedthru by four teflon coated constantan wires. The wires shall be less than six inches in length and the resistance shall be less than one percent of the value of the resistor in the coldstation.

5. The coldstation is clamped to the coldfinger, sufficiently tight to effect good heat transfer without damaging the coldfinger.

6. Good thermal conductivity is assured by applying a metal oxide impregnated silicone thermal joint compound to the end of the coldfinger that is in contact with the coldstation.

7. A radiation shield shall not be used.

4.6.2 Cooldown time. Cooldown time shall be measured using the test Dewar and the copper mass described in 4.6.1. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Input power. Input power shall be measured when the cold tip temperature has stabilized below 80 K with heat load applied. Temperature stabilization has occurred when the output of the temperature sensor diode does not change more than .015 Vdc over a 5 minute period. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

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4.6.4 Acoustic noise. The cooler shall be set up for operation in an area where the background noise level is at least 5 dB below the sound level to be measured. The cooler shall be operated and sound pressure measurements made with the cooler oriented in 4 positions in the same plane with each position approximately 90 degrees apart. Measurements shall be made with an octave-band analyzer with characteristics which comply with ANSI Specification S1.11-1971. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Leak rate. The cooler shall be placed in a bell jar or suitable fixture connected to a helium mass spectrometer. The bell jar or fixture shall be evacuated to establish an inside-out test mode. Operating test shall not be performed during leak rate test. The leakage rate shall be measured at +23°C, ±5°C ambient. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Vibration output. The cooler shall be turned on for 15 minutes to allow it to reach equilibrium temperature. The cooler shall be vertically suspended such that the fundamental frequency of the cooler is much higher than the natural frequency of the suspension system. The static deflection of this arrangement shall be in excess of 6 inches and critical frequency (f_{cr}) of 1.3 Hz approximately. The acceleration along each axis shall be obtained by attaching an accelerometer to the center of the piston and motor faces and to the unit centered on the coldfinger axis. Moments around each axis shall be obtained by attaching two accelerometers (to the motor, piston face or coldfinger face) and coherently subtracting the two signals. This difference divided by the separation distance between the accelerometers yields the angular acceleration. Moment of inertia shall be obtained from the period of a bifilar pendulum containing the cooler. Peak forces are derived from the expression F = MA, where F is the peak force. M is the cooler mass, and A is the measured acceleration at a given frequency. Peak torque is derived from the expression $T = I\ddot{\theta}$, where T is the peak torque, I is the moment of inertia about the principal axis, and $\ddot{\theta}$ is the measured angular acceleration for the applicable axis at a given frequency. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Restart. The cooler shall be operating at 80K or less in a 23C ± 5C ambient. The power switch shall be turned off and the cooler motor shall be at rest. The power switch shall be turned on and the cooler must start within 5 seconds. This test shall be repeated a total of twenty times. The cooler shall also be tested at -54C and +71C using the above procedure during the high and low tests of 4.7. The restart test shall be performed after the cooldown and cooling capacity tests. Failure to meet the requirement of 3.6.7 shall constitute failure of this test.

4.7 Environmental tests. Unless otherwise specified, the environmental tests shall be performed in accordance with MIL-STD-810. The operating tests required before and after environmental tests shall be those specified in table III. During 4.7.2 and 4.7.3, operating test 4.6.1, 4.6.2, 4.6.3 and 4.6.7 shall be performed and must be successfully completed. Operating tests are not required during 4.7.1, 4.7.4 and 4.7.5 except that the unit under test shall have power applied during 4.7.4 (Procedure IV only) and 4.7.5.

4.7.1 Temperature shock. The cooler shall be tested in accordance with MIL-STD-810, Method 503.1, Procedure I except that the temperature of step 2 shall be -54°C and the temperature of step 1 and step 4 shall be +95°C. Failure to meet requirements of 3.7.1 shall constitute failure of this test.

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4.7.2 High temperature. The cooler shall be tested in accordance with MIL-STD-810, Method 501.1, Procedure I (see figure 6) except that the temperature of step 2 shall be +95°C. The temperature of steps 4 and 5 shall be +71°C. Failure to meet requirements of 3.7.2 shall constitute failure of this test.

4.7.3 Low temperature. The cooler shall be tested in accordance with MIL-STD-810, Method 502.1, Procedure I (see figure 7) except that the storage temperature of step 2 shall be -62°C and low operating temperature of step 4 shall be -54°C. Failure to meet requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Shock. The cooler shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure IV, figure 516.2-1 (100 g's peak value at 11 milli-seconds time duration) and Procedure V. Failure to meet requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Vibration. The cooler shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure VIII except that the curve shown in figure 4 of this specification shall be used in lieu of the MIL-STD-810 curve. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be 1/6 of the cycling time at each resonance. Failure to meet requirements of 3.7.5 shall constitute failure of this test.

4.8 Burn-in. Each cooler shall be subjected to a minimum of 12 hours burn-in in accordance with figure 5. Failure of burn-in shall be defined as failure to pass the inspections listed in table III after burn-in. Any unit which fails and is repaired or is later altered, repaired or reworked, shall be resubjected to burn-in.

4.9 Reliability.

4.9.1 Reliability. The lower MTTF shall be demonstrated using 4 coolers in accordance with the reliability test cycle shown in figure 8. Heat load, input power, cold tip temperature and ambient temperature shall be monitored hourly. Cooldown time shall be measured twice during each cycle of figure 8, at -32°C and +52°C. The tests of paragraph 4.9.1.1 shall be performed at least once prior to and once after the reliability test. The results of all 4.9.1.1 tests shall be used to determine the success or failure of the reliability test.

4.9.1.1 Failure definition. Failure shall be defined by any one of the four criteria below:

a. Inability to cool at least 90% of heat load carrying capability in accordance with figure 1, curve C at any ambient measured at 90 K when measured at any point during the 40-hour reliability cycle.

b. Failure to cooldown to 90 K in 15 minutes.

c. Failure to meet 55 watts input power when measured at any point during the 40-hour reliability cycle.

d. Failure to meet the leak rate requirement of 3.6.5.

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Should a failure occur in one of the four categories above, the hours accumulated since completion of the previously successful test, shall not be included in the accept/reject decision.

4.9.1.2 Accept/Reject decision. The accept/reject decision shall be made by the Government in accordance with the following definition. Four coolers shall be run until 4000 total "on" hours have been accumulated. A cooler which experiences a relevant failure as determined by the Government shall not be repaired and put back into test. No single unit "on" time shall be less than ½ the average operating time of all units on test and no single unit "on" time shall be greater than 1.25 the average operating time of all units on test.

4.10 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 10-1-1-00-NS-X-ED-0-00-A.

5.1.2 Level B. 10-1-1-00-NS-X-ED-0-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The cooler is intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required except for existing suppliers (6.3.2).

1. Time frame for submission of first article test reports when first article tests are required.

2. Time frame for approval of first article test reports when first article tests are required.

c. Production delivery schedule.

1. Defined in terms of monthly lots.

2. Actions required relative to group C failures.

3. Actions required relative to group D failures.

d. Level A or level B preservation and packaging (the coldfinger is extremely delicate; packaging shall protect it from bending or distortion)(see section 5).

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e. MIL-STD-810C shall be used for environmental tests of 4.7.

f. Deliver all first article and Group D samples to - Commander, USACECOM, ATTN: AMSEL-NV-CM, Fort Belvoir, VA 22060.

g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specification or preparing activity for recommended disposability methods.

h. Group C and Group D inspection sample size to be established based on cumulative totals of concurrent contracts.

i. Suppliers are those who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12 month period. If a supplier cannot maintain the minimum monthly average, a first article test (less reliability) performed annually shall sustain the existing supplier's status when performed on 3 units manufactured by the supplier explicitly for this test. Upon implementation of this specification, suppliers who are then supplying equipment to specification B2-28A050108, Cooler, Cryogenic, Mechanical HD-1033/UA, HD-1033A/UA with minimum monthly delivery rates specified above, shall be considered as existing suppliers.

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of cooler performance to values less than specified herein.

6.3.2 Lower mean time to failure. The term "Lower" is used to define that value which is the least operational capability and/or the maximum operating and support cost burden the army can tolerate and accept. This distinction is made to aid the system designer in allocating reliability characteristics of the device.

6.4 Design note. Cooler input power will increase when tested with a Dewar (0.4 WHL) from 50 w to 53 w at 23°C ambient temperature and from 55 w to 60 w at 71°C ambient temperature.

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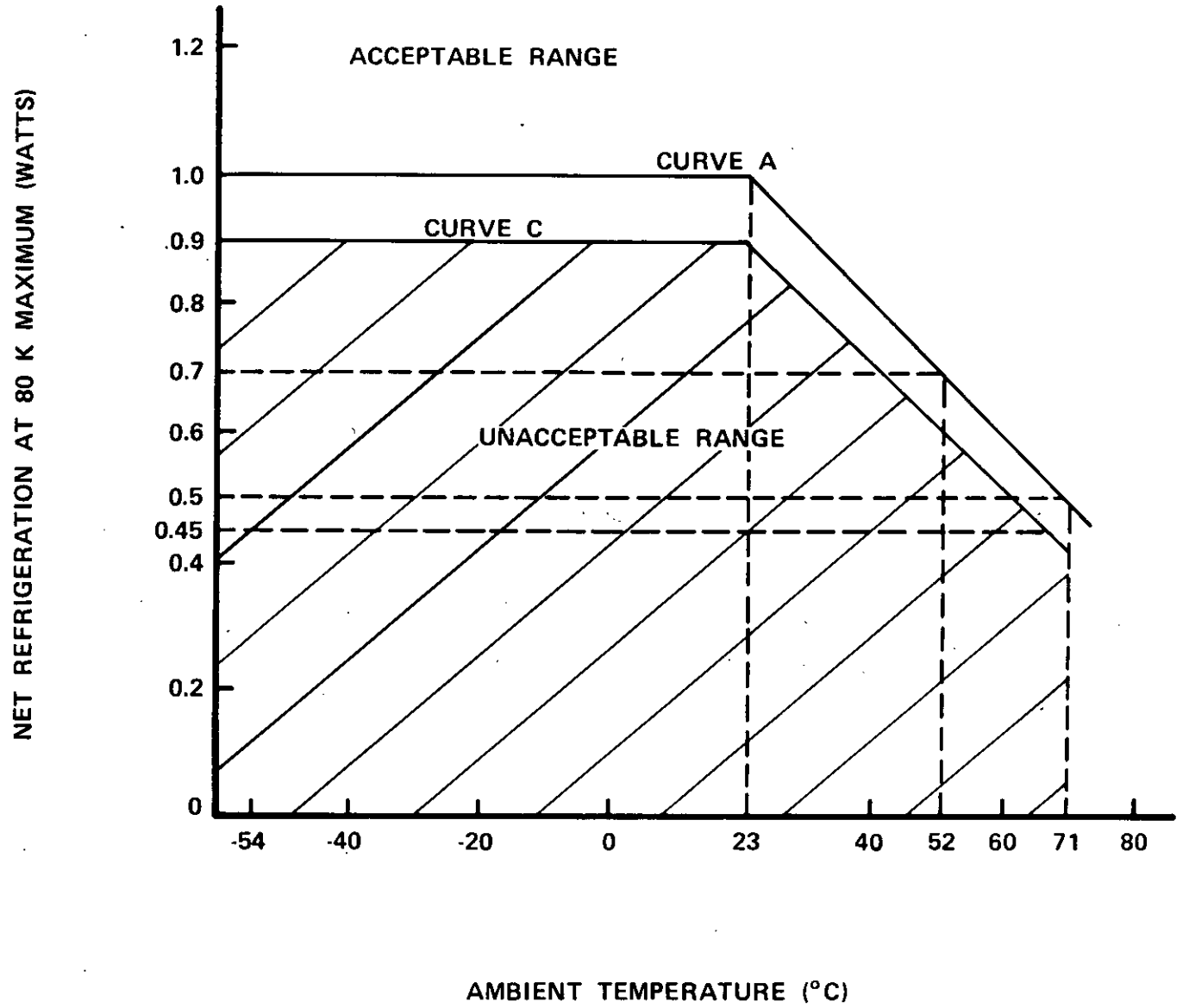


FIGURE 1. Cooling capacity.

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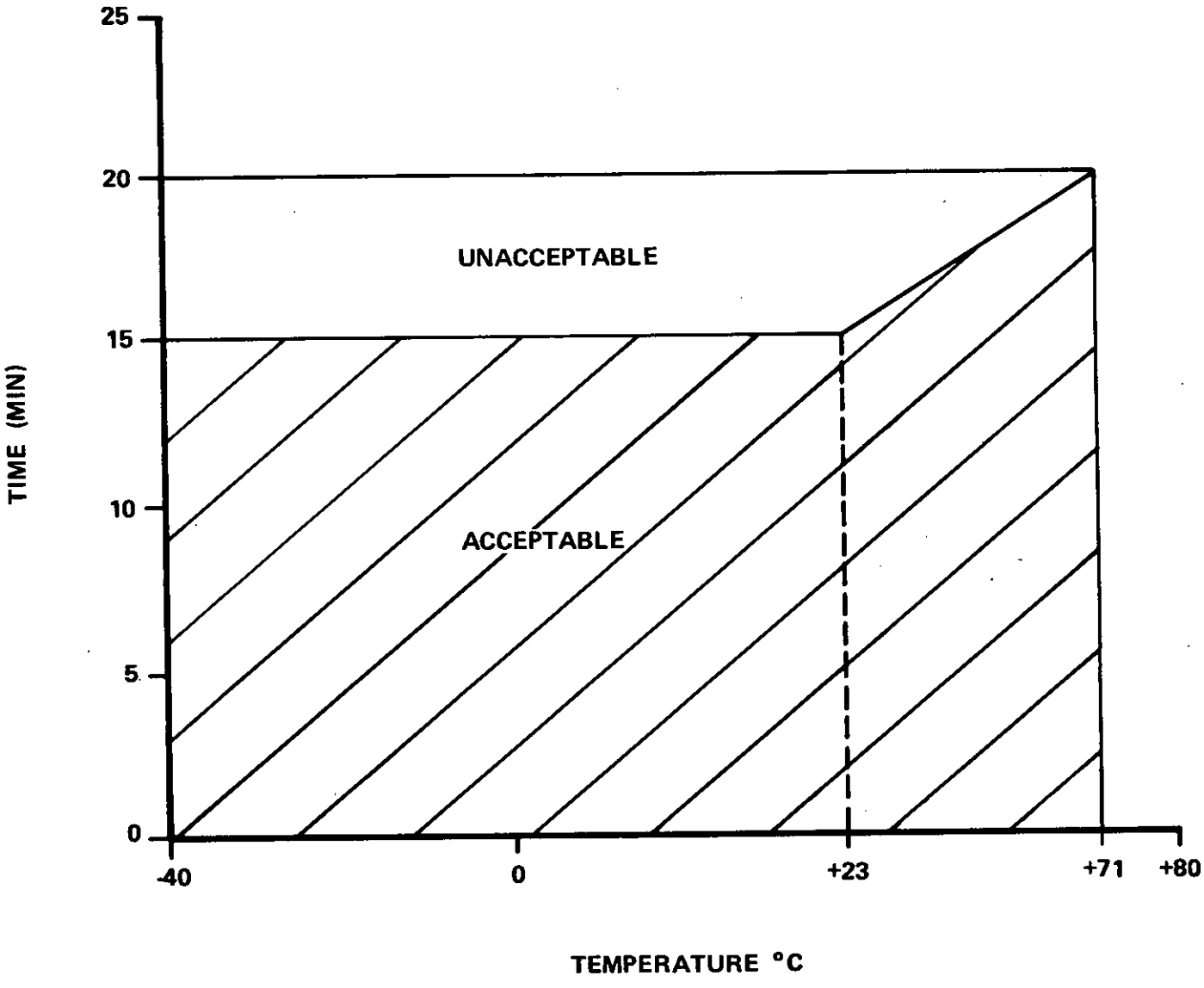


FIGURE 2. Cool down curve.

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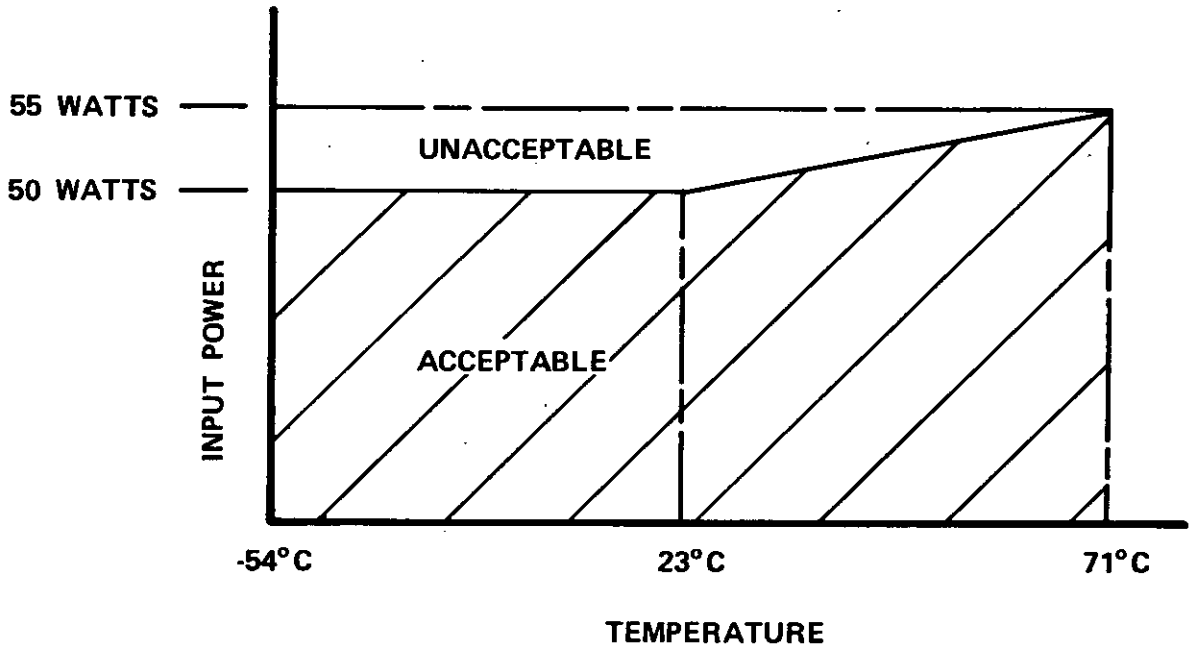


FIGURE 3. Input power with heat load applied.

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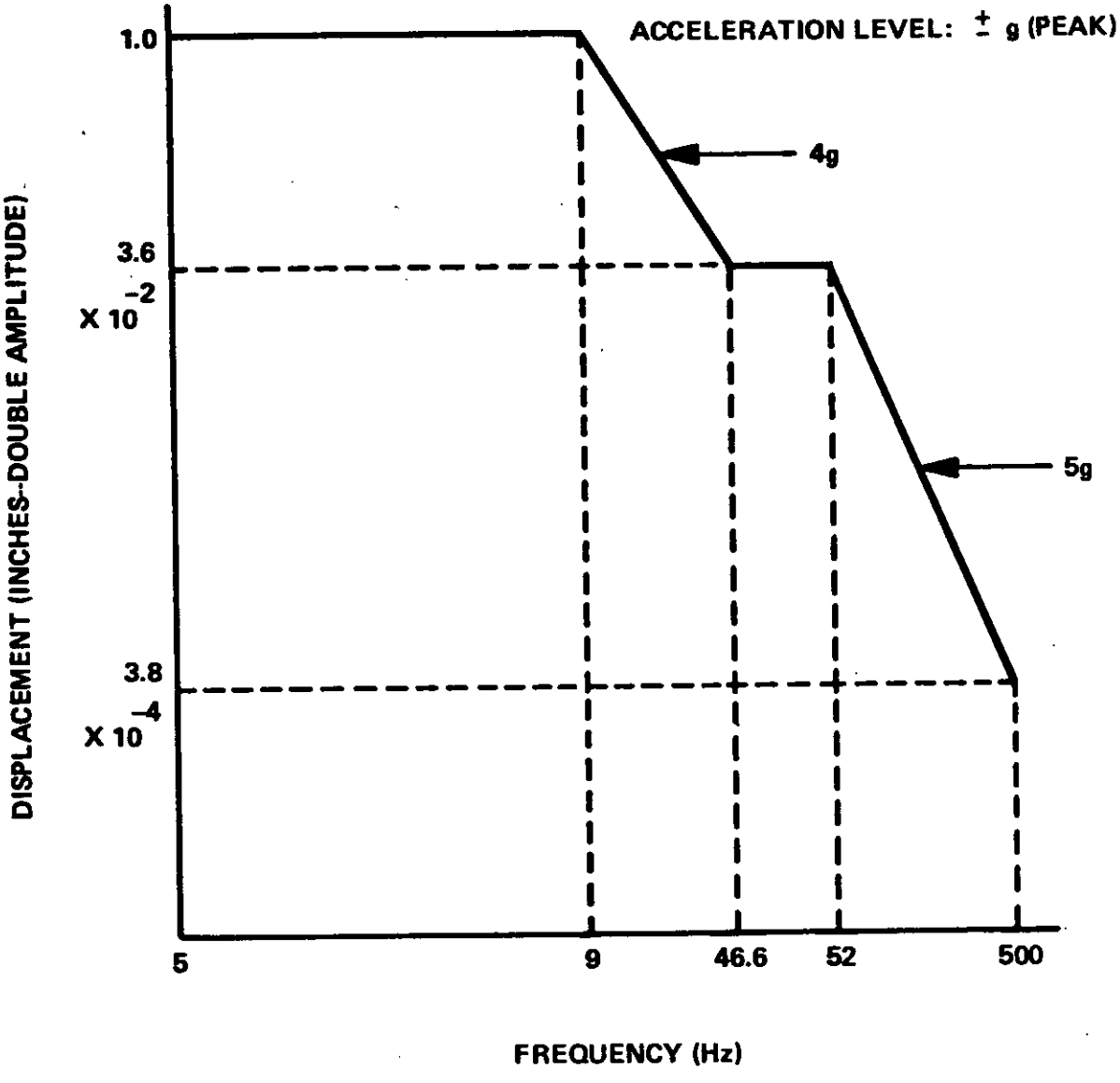
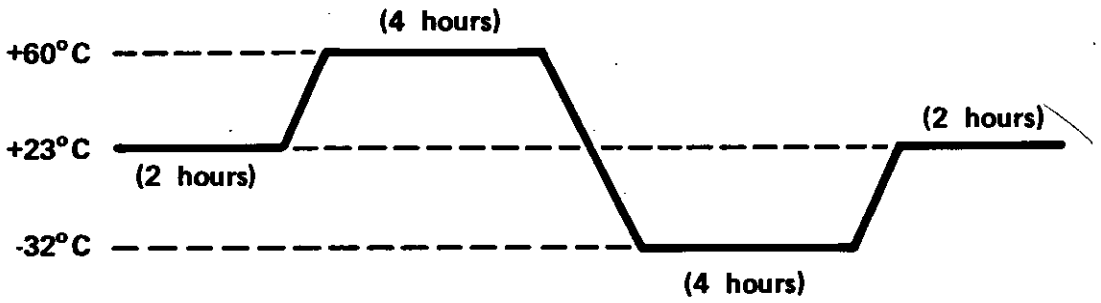


FIGURE 4. **VIBRATION TEST PROFILE**

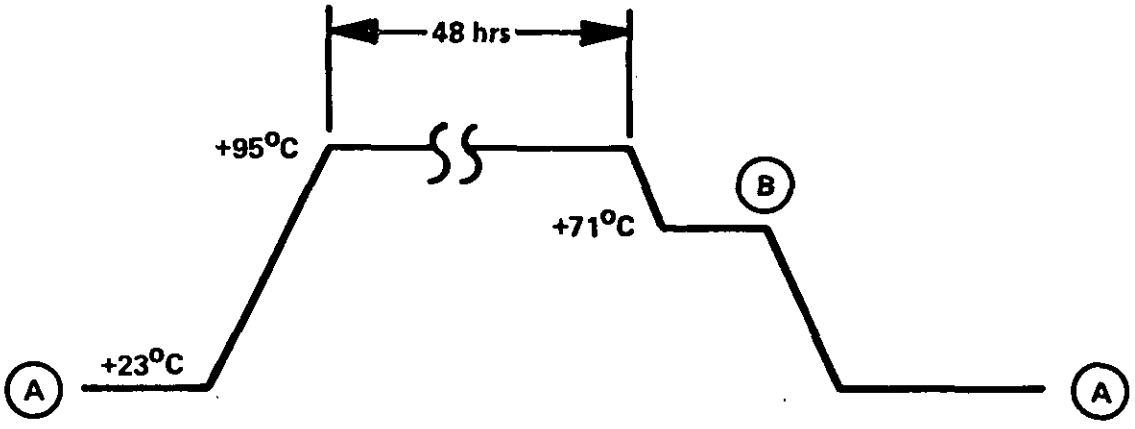
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1) CHAMBER TEMPERATURE EXTREMES SHALL NOT VARY MORE THAN 5°C PER HOUR.
2) RATE OF CHAMBER TEMPERATURE CHANGE SHALL AVERAGE LESS THAN 5°C PER MINUTE.
3) UNIT OPERATES THROUGH FULL CYCLE.

FIGURE 5. Burn-in.

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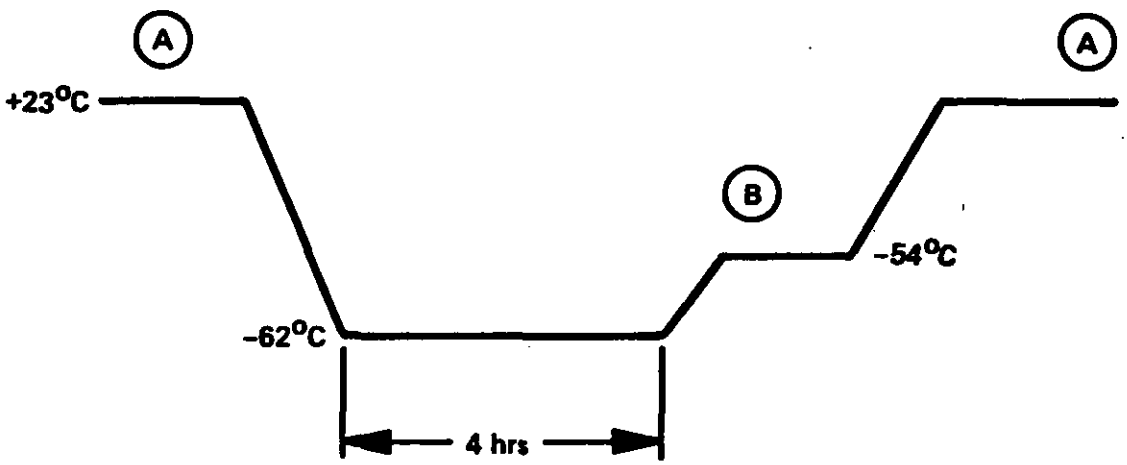


(A) GROUP A TESTS (TABLE III) ARE CONDUCTED AT THESE POINTS.

(B) COLDFINGER TEMPERATURE WITH 0.5 WATT APPLIED HEAT LOAD MONITORED AT THIS POINT TO VERIFY ABILITY OF COOLER TO MAINTAIN 80 K TEMPERATURE.

FIGURE 6. **HIGH TEMPERATURE TEST PROFILE**

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(A) GROUP A TESTS (TABLE III) ARE CONDUCTED AT THESE POINTS.

(B) COLDFINGER TEMPERATURE WITH 0.8 WATT APPLIED HEAT LOAD MONITORED AT THIS POINT TO VERIFY ABILITY OF COOLER TO MAINTAIN 80 K TEMPERATURE.

FIGURE 7. **LOW TEMPERATURE TEST PROFILE**

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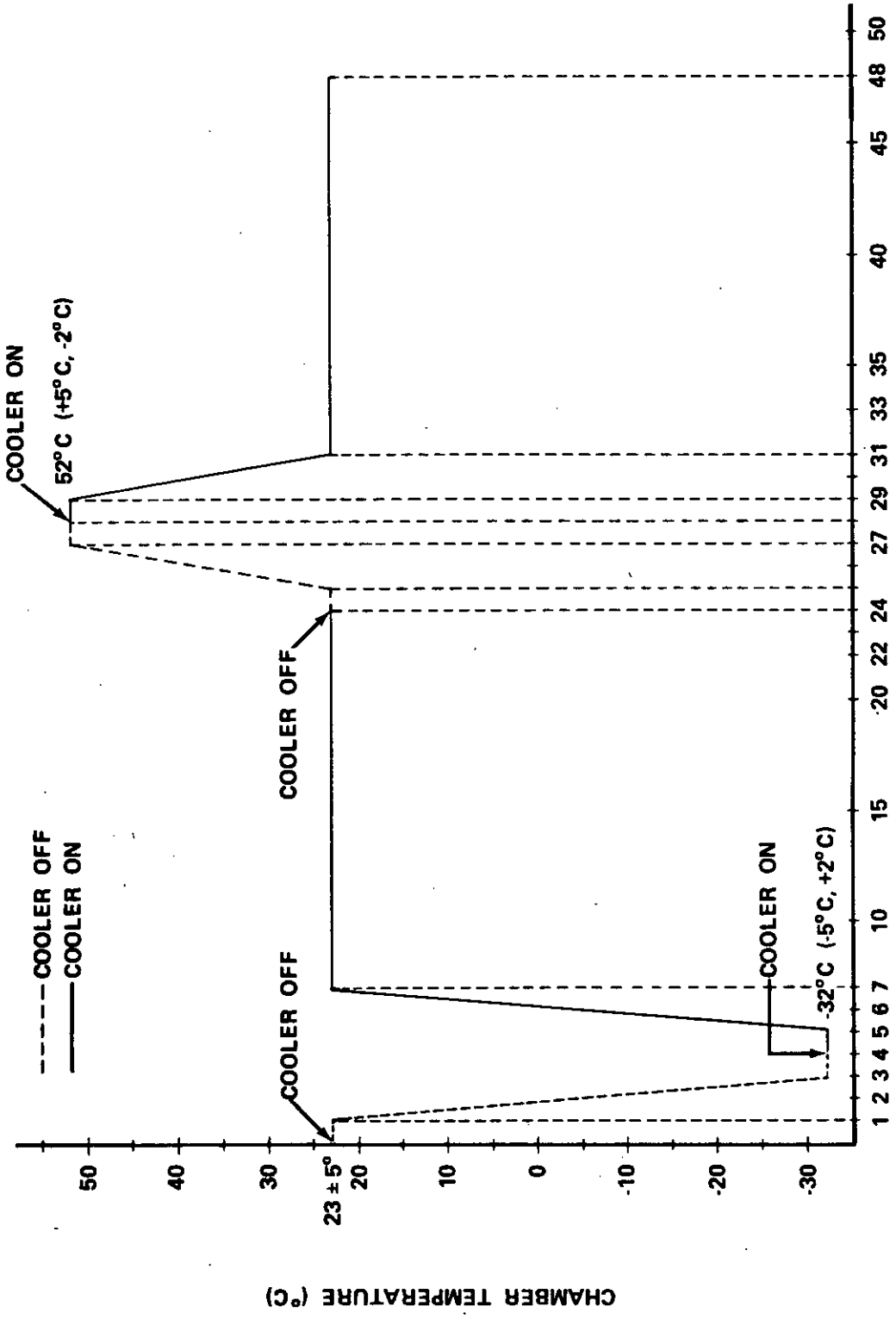


FIGURE 8. Life test.

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